

```
)

return output_text.split("Answer: ")[-1].strip()

while True:
    question = input("\nEnter your question (or type 'exit'): ")
    if question.lower() == "exit":
        break

    answer = generate_rag_answer(question)
    print("\nAnswer:", answer)
```

```
Enter your question (or type 'exit'): what are the ai technique used
in ai??
```

Answer: machine learning, data-driven approaches, and data-driven prognostics.

Enter your question (or type 'exit'): what is prognostic

Answer: Prognostic is a term used to describe the process of predicting the remaining useful life of a system or component. Prognostics is a field of engineering that involves the development of algorithms and models to predict the remaining useful life of a system or component. Prognostics is used in various industries such as aerospace, automotive, and defense. Prognostics is a field of engineering that involves the development of algorithms and models to predict the remaining useful life of a system or component. Prognostics is used in various industries such as aerospace, automotive, and defense. Prognostics is a field of engineering that involves the development of algorithms and models to predict the remaining useful life of a system or component. Prognostics is used in various industries such as aerospace, automotive, and defense. Prognostics is a field of engineering that involves the development of algorithms and models to predict the remaining useful life of a system or component. Prognostics is used in various industries such as aerospace, automotive, and defense.

```
Enter your question (or type 'exit'): exit
```

Experiment 8

```
!pip install transformers datasets evaluate sentencepiece sacrebleu rouge-score
```

Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)

```
Requirement already satisfied: datasets in
/usr/local/lib/python3.12/dist-packages (4.0.0)
Collecting evaluate
  Downloading evaluate-0.4.6-py3-none-any.whl.metadata (9.5 kB)
Requirement already satisfied: sentencepiece in
/usr/local/lib/python3.12/dist-packages (0.2.1)
Collecting sacrebleu
  Downloading sacrebleu-2.6.0-py3-none-any.whl.metadata (39 kB)
Collecting rouge-score
  Downloading rouge_score-0.1.2.tar.gz (17 kB)
  Preparing metadata (setup.py) ... ent already satisfied: filelock in
/usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (1.7.1)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in
/usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in
/usr/local/lib/python3.12/dist-packages (from transformers)
(2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer-slim in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in
/usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in
/usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: pyarrow>=15.0.0 in
/usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in
/usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in
/usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in
/usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: xxhash in
/usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocessing<0.70.17 in
/usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in
/usr/local/lib/python3.12/dist-packages (from
fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (2025.3.0)
Collecting portalocker (from sacrebleu)
  Downloading portalocker-3.2.0-py3-none-any.whl.metadata (8.7 kB)
Requirement already satisfied: tabulate>=0.8.9 in
```

```
/usr/local/lib/python3.12/dist-packages (from sacrebleu) (0.9.0)
Collecting colorama (from sacrebleu)
  Downloading colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Requirement already satisfied: lxml in /usr/local/lib/python3.12/dist-packages (from sacrebleu) (6.0.2)
Requirement already satisfied: absl-py in
/usr/local/lib/python3.12/dist-packages (from rouge-score) (1.4.0)
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (from rouge-score) (3.9.1)
Requirement already satisfied: six>=1.14.0 in
/usr/local/lib/python3.12/dist-packages (from rouge-score) (1.17.0)
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in
/usr/local/lib/python3.12/dist-packages (from
fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (3.13.3)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (1.4.2)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (0.28.1)
Requirement already satisfied: typer in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (0.24.1)
Requirement already satisfied: typing-extensions>=4.1.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (4.15.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (3.4.6)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.2-
>datasets) (2026.2.25)
Requirement already satisfied: click in
/usr/local/lib/python3.12/dist-packages (from nltk->rouge-score)
(8.3.1)
Requirement already satisfied: joblib in
/usr/local/lib/python3.12/dist-packages (from nltk->rouge-score)
(1.5.3)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets)
```

(2025.2)

Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.12/dist-packages (from pandas->datasets)

(2025.3)

Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (2.6.1)

Requirement already satisfied: aiosignal>=1.4.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.4.0)

Requirement already satisfied: attrs>=17.3.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (25.4.0)

Requirement already satisfied: frozenlist>=1.1.1 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.8.0)

Requirement already satisfied: multidict<7.0,>=4.5 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (6.7.1)

Requirement already satisfied: propcache>=0.2.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (0.4.1)

Requirement already satisfied: yarl<2.0,>=1.17.0 in
/usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,
=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.23.0)

Requirement already satisfied: anyio in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (4.12.1)

Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (1.0.9)

Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.16.0)

Requirement already satisfied: shellingham>=1.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (1.5.4)

Requirement already satisfied: rich>=12.3.0 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (13.9.4)

Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (0.0.4)

Requirement already satisfied: markdown-it-py>=2.2.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (4.0.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (2.19.2)

```

Requirement already satisfied: mdurl~=0.1 in
/usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0-
>rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers)
(0.1.2)
Downloading evaluate-0.4.6-py3-none-any.whl (84 kB)
_____ 84.1/84.1 kB 3.8 MB/s eta
0:00:00
_____ 100.8/100.8 kB 10.9 MB/s eta
0:00:00
a-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading portaloocker-3.2.0-py3-none-any.whl (22 kB)
Building wheels for collected packages: rouge-score
  Building wheel for rouge-score (setup.py) ... e=rouge_score-0.1.2-
py3-none-any.whl size=24934
sha256=4d9dbcee2b1fab702141a3d730fad4a2453be2141c595c9bce06788c3151f92
1
  Stored in directory:
/root/.cache/pip/wheels/85/9d/af/01feefbe7d55ef5468796f0c68225b6788e85
d9d0a281e7a70
Successfully built rouge-score
Installing collected packages: portaloocker, colorama, sacrebleu,
rouge-score, evaluate
Successfully installed colorama-0.4.6 evaluate-0.4.6 portaloocker-3.2.0
rouge-score-0.1.2 sacrebleu-2.6.0

import torch
import pandas as pd
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
from datasets import Dataset
from sklearn.model_selection import train_test_split
from evaluate import load

df = pd.read_csv("bank_faq.csv")

print(df.head())    # check data

      question
answer
0      What is ATM?  ATM is a machine used to withdraw cash
from ba...
1      What is net banking?  Net banking allows customers to perform
bankin...
2      What is mobile banking?  Mobile banking allows banking services
through...
3      What is IFSC code?  IFSC code identifies a specific bank
branch fo...
4      What is a debit card?  A debit card allows you to access money
from y...

```

```
train_q, test_q, train_a, test_a = train_test_split(
    df["question"], df["answer"], test_size=0.3, random_state=42
)
```

```
train_dataset = Dataset.from_dict({
    "question": list(train_q),
    "answer": list(train_a)
})
```

```
test_dataset = Dataset.from_dict({
    "question": list(test_q),
    "answer": list(test_a)
})
```

```
model_name = "google/mt5-small"
```

```
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/
_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your
settings tab (https://huggingface.co/settings/tokens), set it as
secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to
access public models or datasets.
warnings.warn(
```

```
{"model_id": "60f557f49f9841eb94da9a5c5d64d1f5", "version_major": 2, "vers
ion_minor": 0}
```

```
{"model_id": "a4c98bf465dd49d7a7d3ca85d450bedd", "version_major": 2, "vers
ion_minor": 0}
```

```
{"model_id": "b919d1df143142049de19a0528e7b3e6", "version_major": 2, "vers
ion_minor": 0}
```

```
{"model_id": "cd6fa3a2091744c98a8f783953c291c1", "version_major": 2, "vers
ion_minor": 0}
```

Warning: You are sending unauthenticated requests to the HF Hub.
Please set a HF_TOKEN to enable higher rate limits and faster
downloads.

WARNING:huggingface_hub.utils._http:Warning: You are sending
unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
enable higher rate limits and faster downloads.

```
{"model_id": "225478ff547f46519cef431970d1c425", "version_major": 2, "vers
ion_minor": 0}
```

```
{"model_id": "3cb587d9849d4aab84eb364d2a52c7df", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "ff7eea9259d2479eb3949feb2111f638", "version_major": 2, "version_minor": 0}
```

The tied weights mapping and config for this model specifies to tie shared.weight to encoder.embed_tokens.weight, but both are present in the checkpoints, so we will NOT tie them. You should update the config with `tie\_word\_embeddings=False` to silence this warning

The tied weights mapping and config for this model specifies to tie shared.weight to decoder.embed_tokens.weight, but both are present in the checkpoints, so we will NOT tie them. You should update the config with `tie\_word\_embeddings=False` to silence this warning

The tied weights mapping and config for this model specifies to tie shared.weight to lm_head.weight, but both are present in the checkpoints, so we will NOT tie them. You should update the config with `tie\_word\_embeddings=False` to silence this warning

```
{"model_id": "c8fcb9e5aba94180bdb47f2148614baf", "version_major": 2, "version_minor": 0}
```

```
def preprocess(batch):
    inputs = tokenizer(batch["question"], padding="max_length",
truncation=True, max_length=64)
    targets = tokenizer(batch["answer"], padding="max_length",
truncation=True, max_length=64)

    inputs["labels"] = targets["input_ids"]
    return inputs
```

```
train_data = train_dataset.map(preprocess, batched=True)
test_data = test_dataset.map(preprocess, batched=True)
```

```
{"model_id": "e9af5913b74f4d1292ff936c859e0acf", "version_major": 2, "version_minor": 0}
```

```
{"model_id": "43b8bb7c12744c03ab32ea1a6a420a20", "version_major": 2, "version_minor": 0}
```

```
optimizer = torch.optim.AdamW(model.parameters(), lr=3e-5)
```

```
model.train()
```

```
for epoch in range(5):
    total_loss = 0

    for batch in train_data:
        optimizer.zero_grad()

        input_ids = torch.tensor(batch["input_ids"]).unsqueeze(0)
```

```

        labels = torch.tensor(batch["labels"]).unsqueeze(0)

        outputs = model(input_ids=input_ids, labels=labels)
        loss = outputs.loss

        loss.backward()
        optimizer.step()

        total_loss += loss.item()

    print(f"Epoch {epoch+1} Loss:", total_loss)

Epoch 1 Loss: 1434.981008529663
Epoch 2 Loss: 1274.0837173461914
Epoch 3 Loss: 1133.052682876587
Epoch 4 Loss: 1091.4537181854248
Epoch 5 Loss: 1039.246810913086

model.eval()

predictions = []
references = []

for batch in test_data:
    input_ids = torch.tensor(batch["input_ids"]).unsqueeze(0)

    outputs = model.generate(input_ids)
    pred = tokenizer.decode(outputs[0], skip_special_tokens=True)

    ref = tokenizer.decode(batch["labels"], skip_special_tokens=True)

    predictions.append(pred)
    references.append(ref)

rouge = load("rouge")
bleu = load("bleu")

rouge_score = rouge.compute(predictions=predictions,
                             references=references)
bleu_score = bleu.compute(
    predictions=predictions,
    references=[[r] for r in references]
)

print("ROUGE-L:", rouge_score["rougeL"])
print("BLEU:", bleu_score["bleu"])

ROUGE-L: 0.00980392156862745
BLEU: 0.0

```



```
def chatbot(question):  
    inputs = tokenizer(question, return_tensors="pt")  
    outputs = model.generate(**inputs)  
    return tokenizer.decode(outputs[0], skip_special_tokens=True)  
  
print(chatbot("why atm used?"))  
  
<extra_id_0>
```